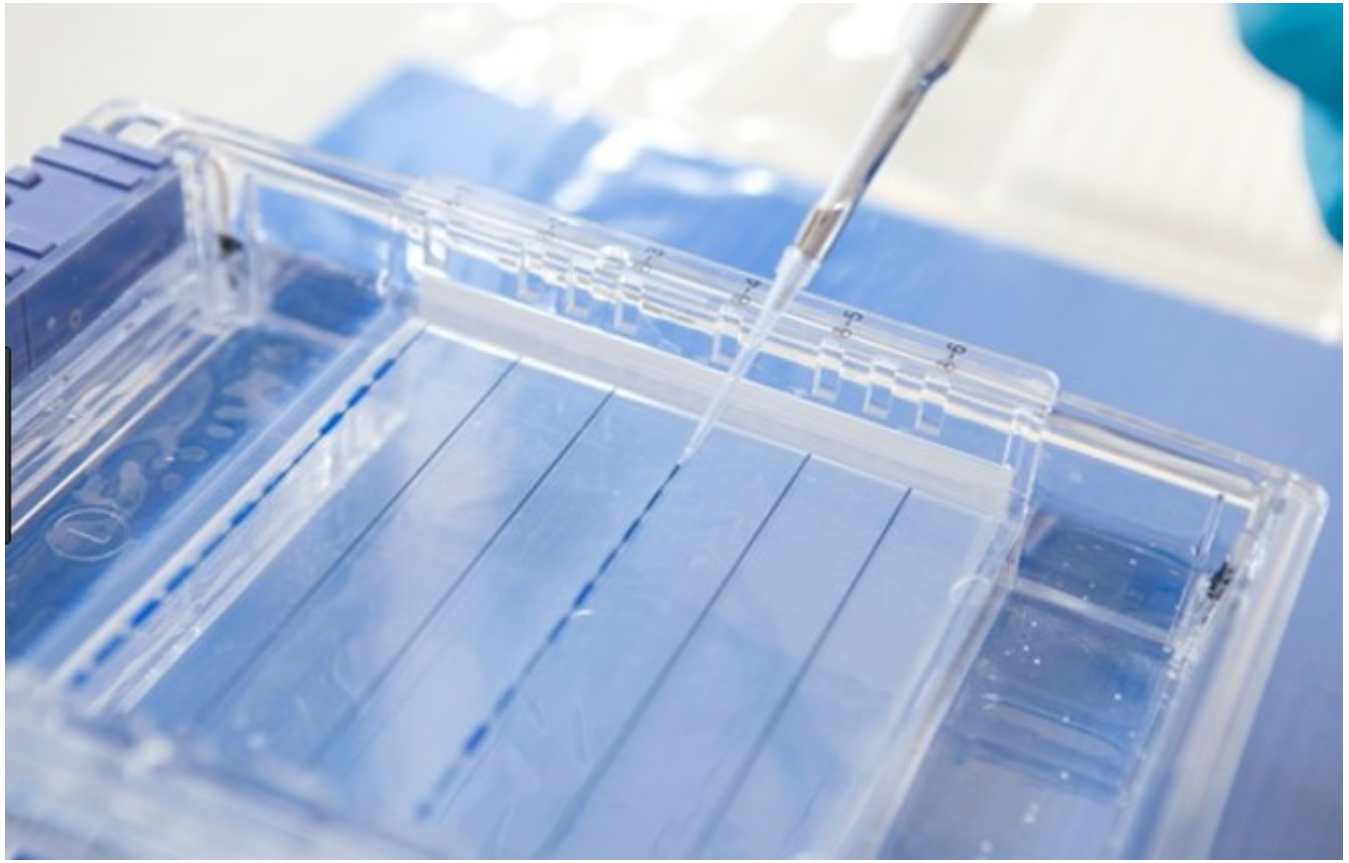
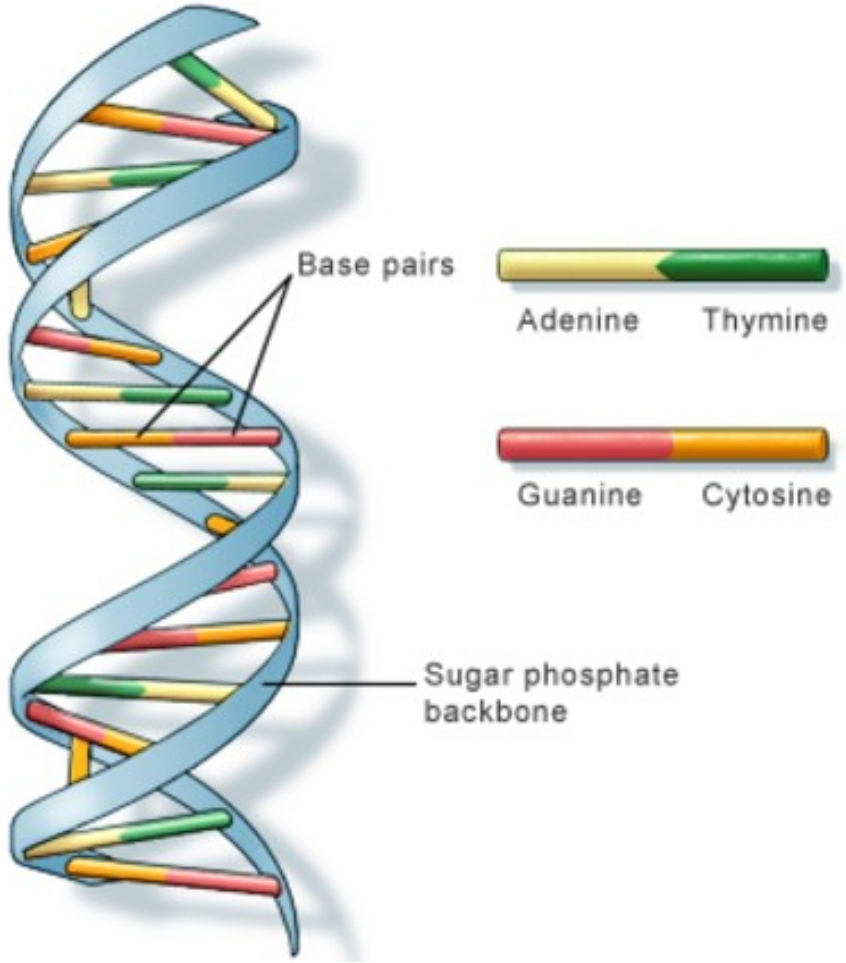
Techniques in Biotechnology



DNA Sequencing and Protein Electrophoresis

DNA reminder

- DNA: Deoxyribonucleic acid
  - Found in cells of all organisms
  - Nucleus and mitochondria
  - Sugar and phosphate backbone
  - Nitrogen base “rungs”
  - 4 bases:
    - Adenine (A)
    - Thymine (T)
    - Cytosine (C)
    - Guanine (G)
  - Order of bases codes for protein production



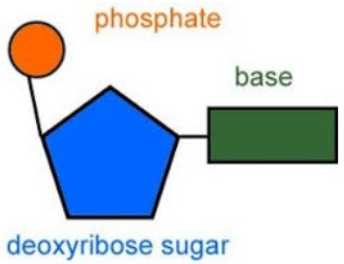
Learning Aim: Review structure of DNA

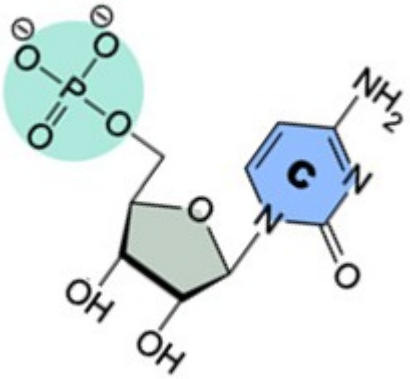
DNA Sequencing

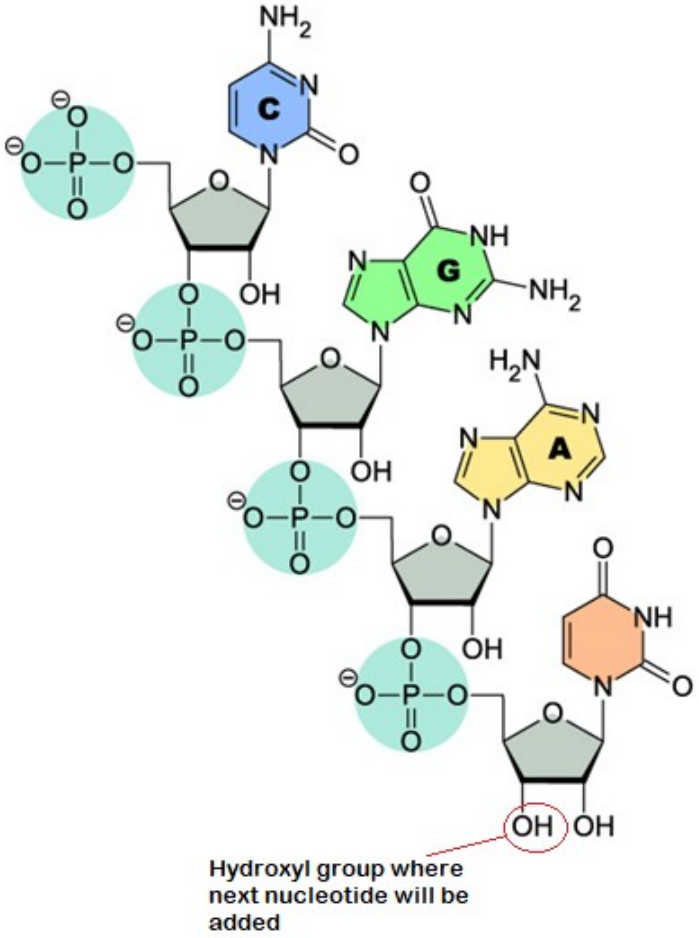
- Phosphate + sugar + base = nucleotide
- During DNA sequencing, DNA polymerase assists in adding nucleotides to a template strand
- As each nucleotide is added, it bonds to the next at the hydroxyl group in the sugar/phosphate backbone. This is an important aspect to remember to understand how the sequencing process works.

- DNA sequencing:
  - Determines order of nucleotide bases
  - Eg CCTTAGATCTTATGCCC…..

- Technique most often used invented by Frederick Sanger – won Chemistry Nobel Prize





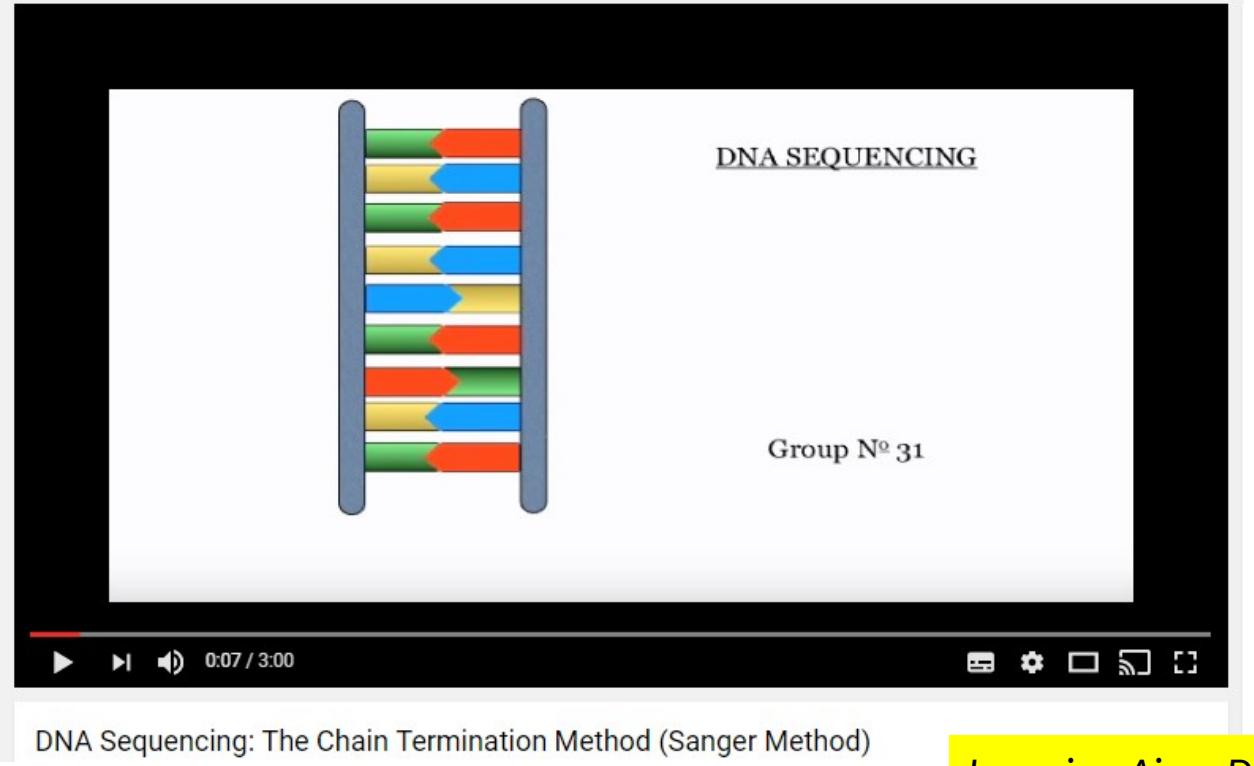


Learning Aim: Review chemical structure of nucleotides and how they are bonded to the sugar/phosphate backbone.

DNA Sequencing: Sanger Technique (aka Chain Termination Method)

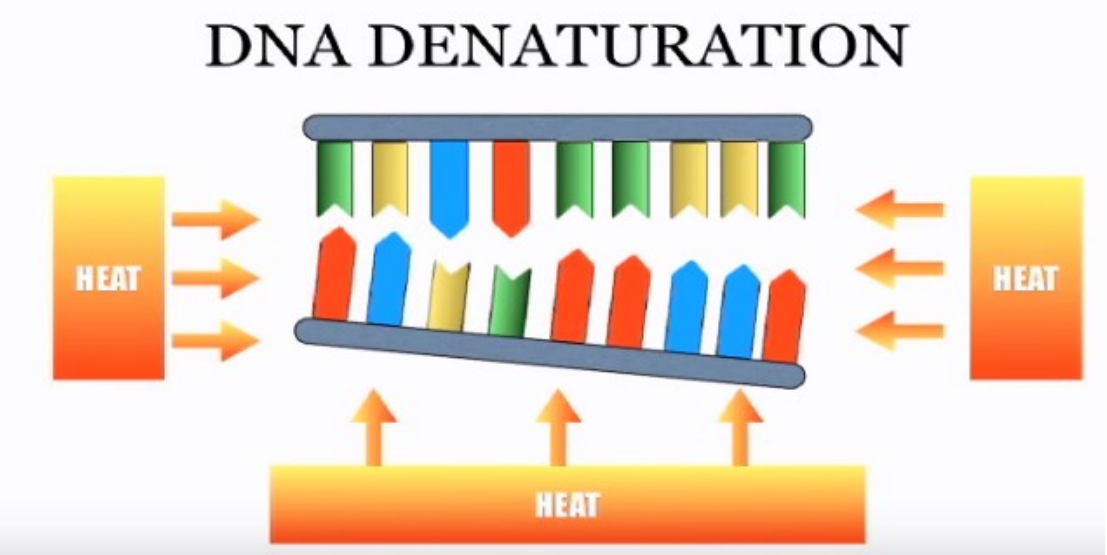
- See Youtube animation at:

https://www.youtube.com/watch?v=vK-HlMaitnE

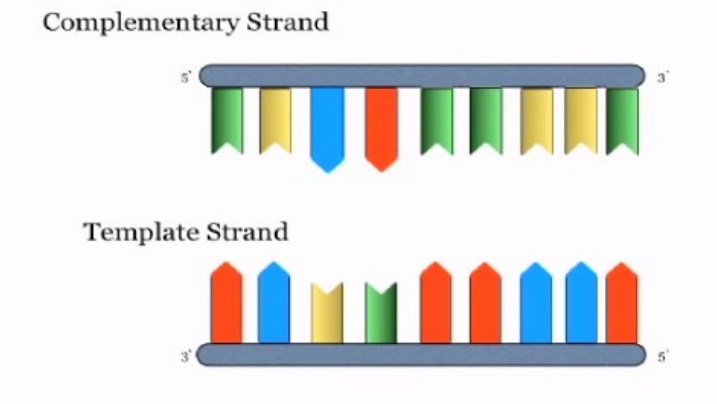


Learning Aim: Describe the steps involved in the Sanger sequencing technique

1: DNA is heated so it denatures into single strands:

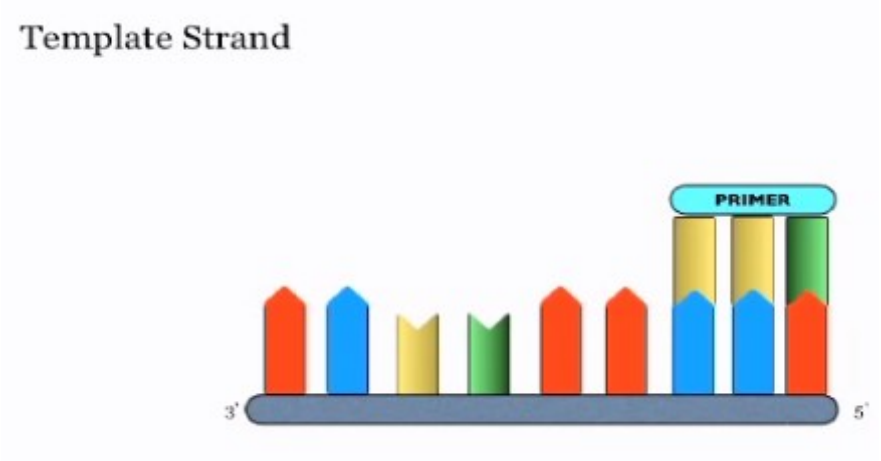


Template strand will be used:



Learning Aim: Describe the steps involved in the Sanger sequencing technique

2: Primer is added to the template strand, to provide a starting point for later nucleotide addition.

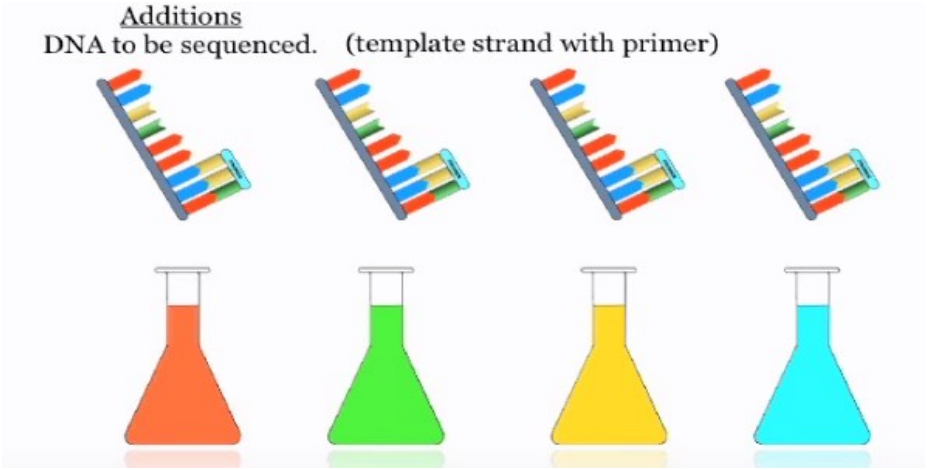


3: Four reaction mixtures are set up. One will be used for each base type.

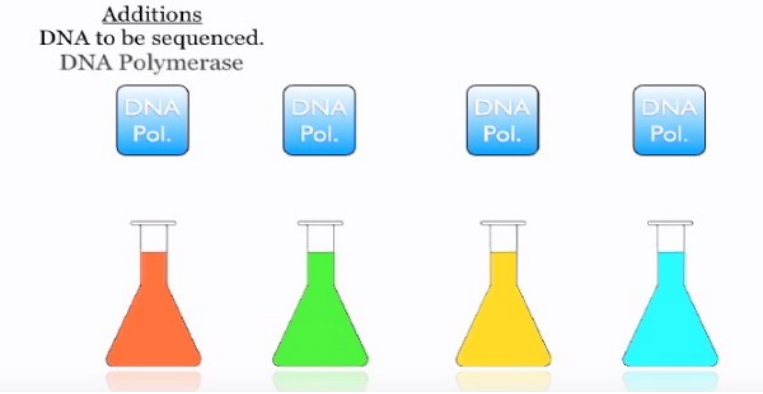


Learning Aim: Describe the steps involved in the Sanger sequencing technique

4: Template strand that will be sequenced, with the annealed primer, is added to each reaction mixture. Many copies are added to each mixture



5: DNA polymerase is added. It will allow free nucleotides to be added to the template strand later.



Learning Aim: Describe the steps involved in the Sanger sequencing technique

6: Free nucleotides (dNTP's) of all types added to each reaction mixture and some radioactively labelled for later identification.

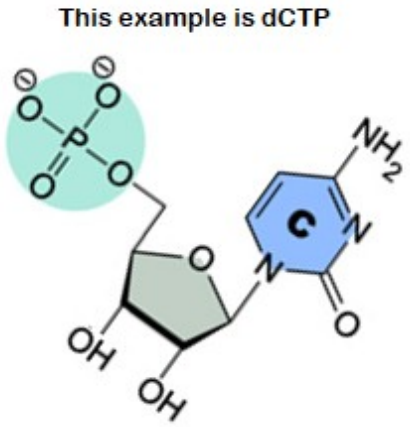
Note: *dNTP = deoxynucleoside triphosphate*

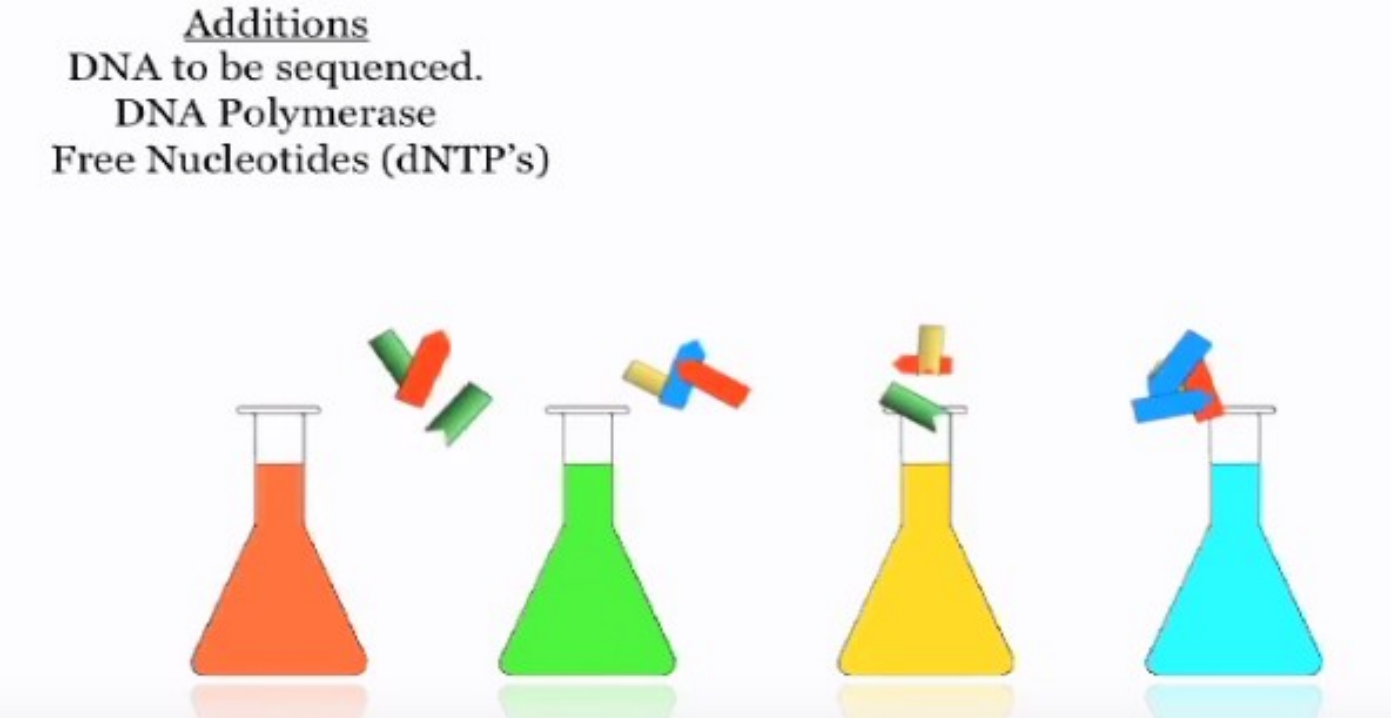
dATP = adenine

dTTP = thymine

dCTP = cytosine

dGTP = guanine



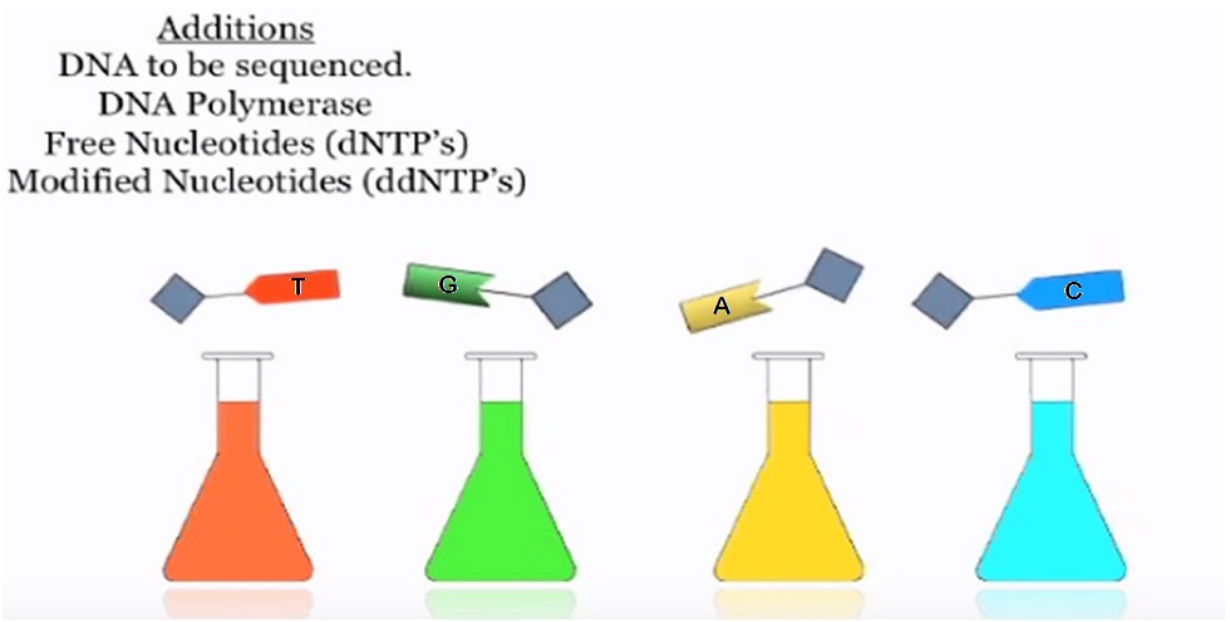


Learning Aim: Describe the steps involved in the Sanger sequencing technique

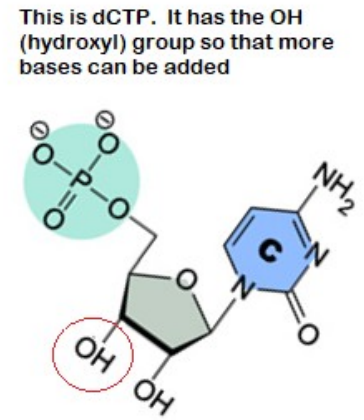
Learning Aim: Describe the steps involved in the Sanger sequencing technique

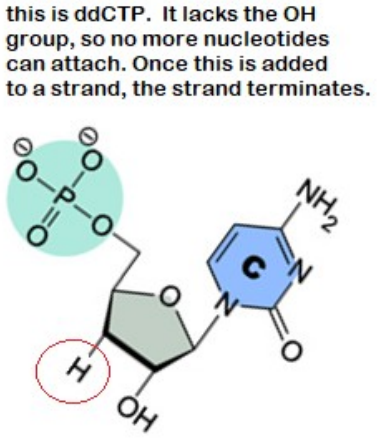
7: Modified nucleotides (ddNTP's – *dideoxynucleotide triphosphate*) added

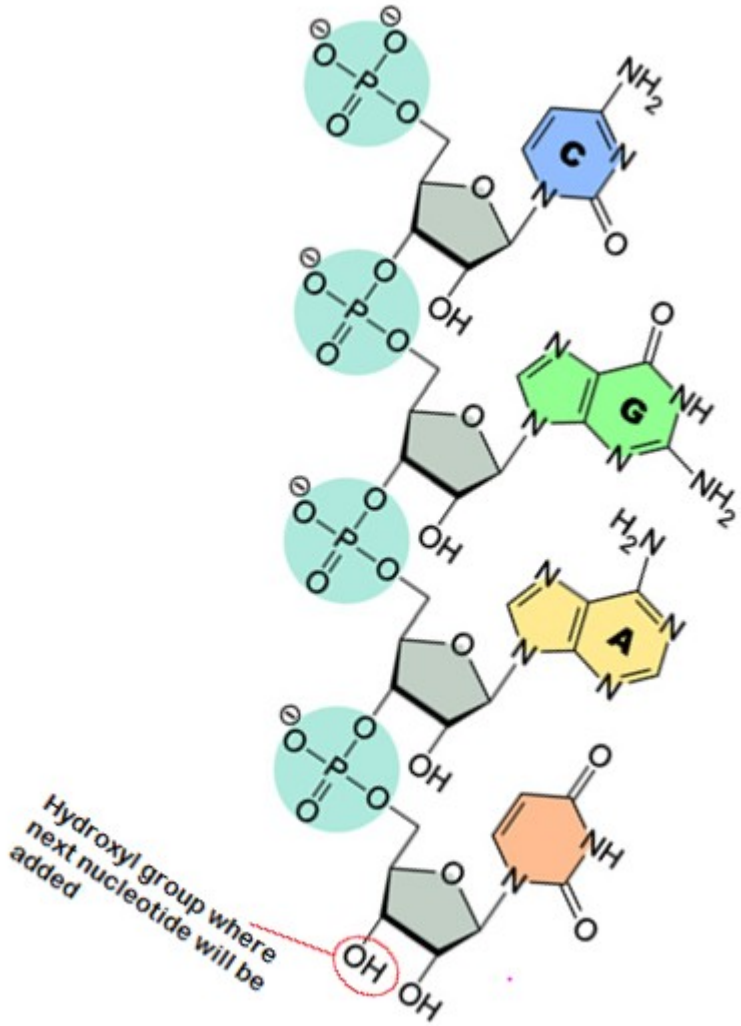
- Only one type is added per reaction mixture
- These lack the hydroxyl group, so no more nucleotides can attach
- This terminates the chain when it is added to the template strand by the DNA polymerase



This means that in the flask with the T added, the template strands will always terminate with a T. They will be different lengths depending on the point at which the ddTTP gets added. So the flask with the T added will end up with lots of different lengths of copied strand, each starting with the primer and ending with a T.

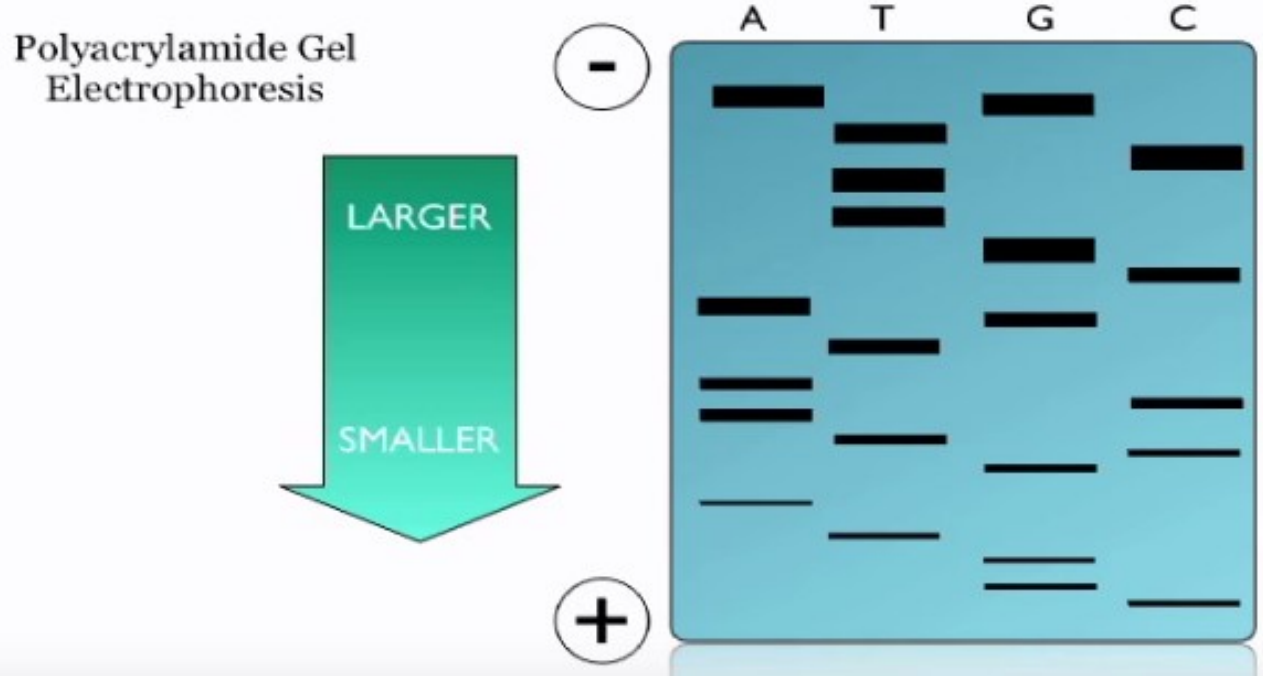






8: Gel electrophoresis used to separate bits of DNA according to size

- One reaction mixture per gel lane
- A current is run through the gel mixture and the negatively charged DNA strands move towards the positive terminal.
- The shortest strands move the fastest
- The strands show up as bands on the gel.
- Can then determine sequence as we know which modified nucleotide terminates each chain



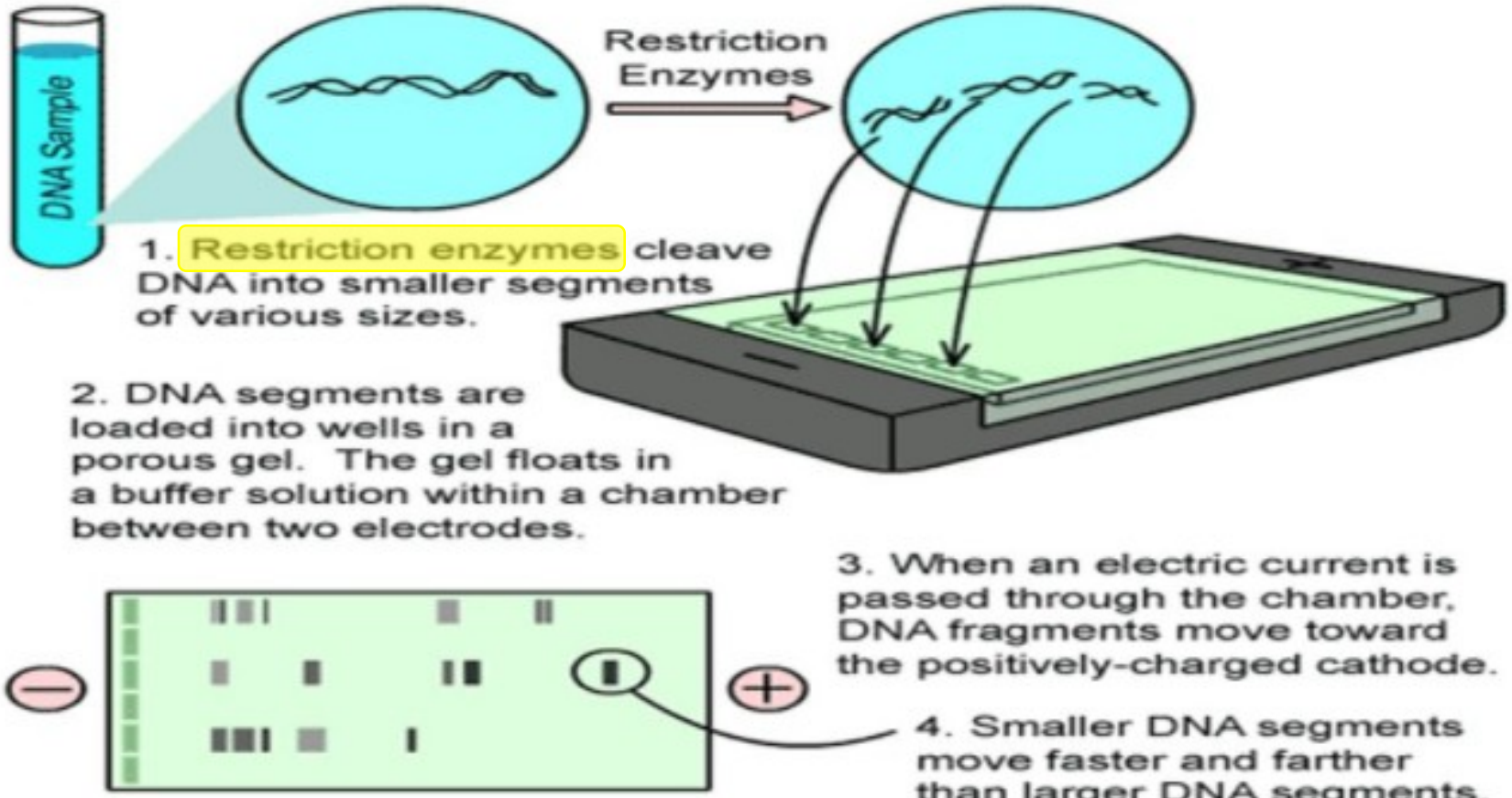
In this case, reading from smallest to largest, we get the sequence:

CGGTAGCTACAT….

Learning Aim: Describe the steps involved in gel electrophoresis

Steps of Gel Electrophoresis

Figure S-2: Gel Electrophoresis



http://www.stanford.edu/group/hopes/diagnsis/gentest/f_s02gelelect.gif

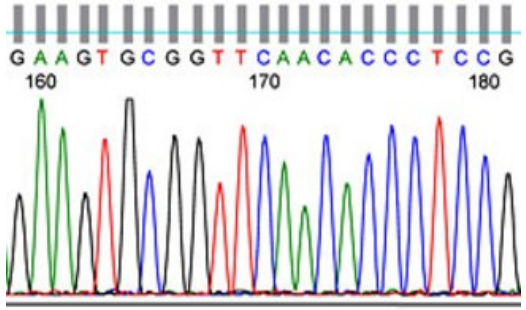
In Sanger Sequencing, it is not restriction enzymes that cut the DNA into segments. It is the ddNTP's that terminate the sequence at different lengths.

Learning Aim: Describe the steps involved in gel electrophoresis

DNA Sequencing: uses

- DNA sequencing can be used to:
  - Compare DNA sequences to detect changed alleles
  - Readily identify point mutations, small insertions and deletions by comparing.
  - Track sequences over time to show evolutionary change.
  - Diseases that can be determined by DNA sequencing:
    - Sickle-cell anaemia
    - Cystic fibrosis
    - Some hereditary forms of cancer

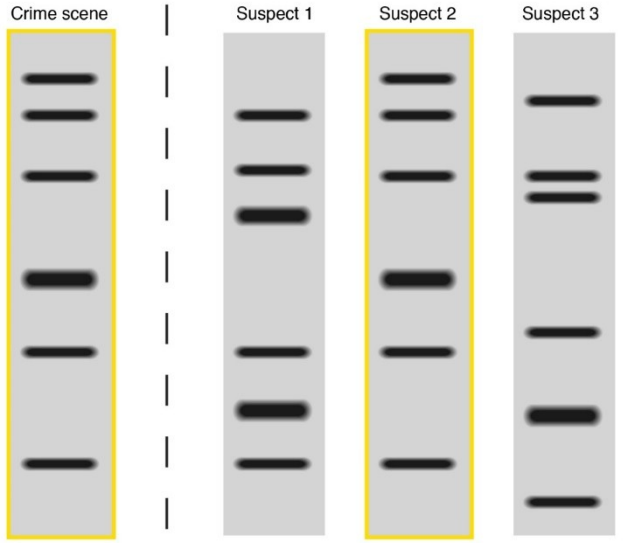
  - Also used for paternity tests and forensic identification – see profiling techniques.

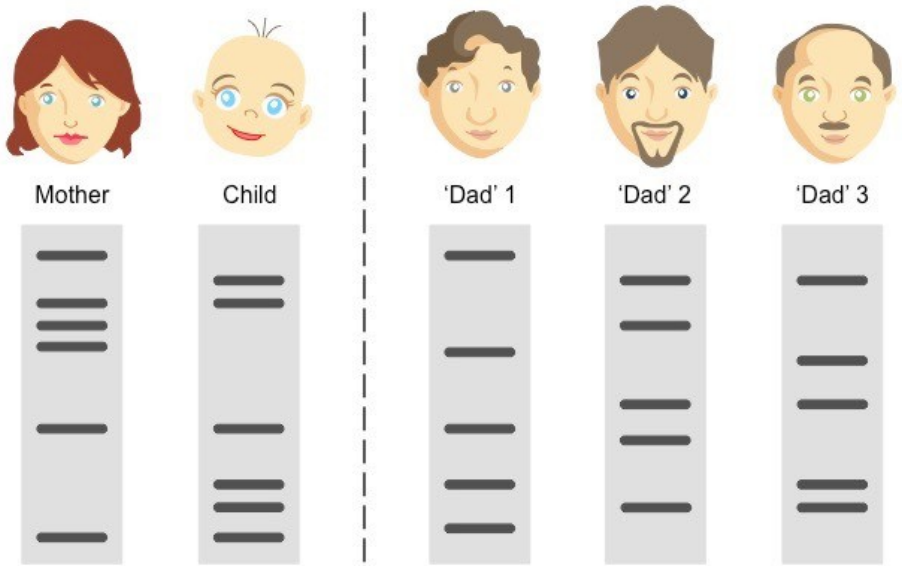


Learning Aim: Discuss uses of DNA sequencing

DNA Profiling Techniques

- DNA can be used as a means of identification
- Specific restriction enzymes cut DNA at specific base sequences, leaving various lengths.
- Lengths vary between individuals as base sequences differ between individuals.
- Gel electrophoresis can show these differences/similarities to determine if two biological samples come from the same person. “DNA fingerprint”.
- Can also be used to determine relatedness between two samples eg paternity testing.
- Can also be used to identify particular gene sequences for some diseases.





Learning Aim: Discuss uses of DNA sequencing

Activity: Modelling Sanger Sequencing

1. Get into groups of 3. Each group will model one of the four reaction mixtures.
2. Gather your equipment. Each person in the group will need:
   - The “template strand for the sequence of interest” on paper. You will need lots of these.
   - A container of “free nucleotides” – counters in different colours.
   - A pen.
   - A pair of scissors.
   - Your teacher will give you a “ddntp terminator base” for the reaction mixture your group is modelling.

3. a. Place a template strand in front of you. Randomly pull a nucleotide from the container, without looking. If it does not bond with the first base after the primer, put it back.

   b. Keep pulling nucleotides out until you get one that matches, then write it in the box and put the counter back.

   c. Repeat for the nucleotides in sequence until a terminator base is added. Write it in the box, then use the scissors to cut the template strand just after the terminator base. Put the strand aside for later.

4. Repeat Step 3 with a new template strand, until you run out of time or template strands.
4. Your teacher will collate the strands for the whole class, and model electrophoresis and reading of the sequence.